

sweeten

A Technical and Research Narrative
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Sweeten is a research-oriented digital health system that investigates how hybrid AI and deterministic architectures can deliver safe, personalized diabetes support under real-world constraints — including API failure, data sparsity, and resource limitations. Built on a failure-first design philosophy, Sweeten remains fully functional and interpretable even when AI components are unavailable. This technical narrative documents the system's architecture, research contributions, clinical grounding, and evaluation strategy.

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1. Background and Motivation

Sweeten was conceived at the intersection of personal observation, public health need, and technological feasibility. Pakistan has over 33 million adults living with diabetes — many undiagnosed or inadequately managed — yet access to structured, personalized health guidance remains limited outside urban centers. Individuals frequently collect health data without the tools to interpret it meaningfully.

Existing digital solutions are either clinically rigid, financially inaccessible, or entirely dependent on continuous AI availability. This last constraint is rarely acknowledged honestly: most health applications assume a stable, well-resourced infrastructure that simply does not reflect deployment conditions in low-resource settings.

Sweeten was designed to address this gap directly — not just as a product, but as a research experiment: what happens to a health-planning system when its AI component fails? Can it remain useful, safe, and personalized?

2. Research Question and Contributions

Core Research Question

Can a health-planning system remain reliable and personalized even when AI components fail — and can the outputs of deterministic and AI-assisted pipelines be meaningfully compared?

Research Contributions

- A hybrid AI + deterministic plan-generation pipeline with graceful degradation
- Structured handling of real-world health data sparsity
- A shared analytics layer enabling direct AI vs. deterministic plan comparison
- ADA guideline-to-code traceability: clinical rules mapped explicitly to system logic
- A reference architecture for AI-resilient health applications in constrained settings
- Ethical constraints on AI usage in healthcare, implemented at the system level

3. My Journey and Skill Development

Sweeten reflects self-directed learning across the full software development stack. Development began with frontend engineering — TypeScript, React, Next.js 15, and Tailwind CSS — to build responsive, accessible interfaces with minimal cognitive load. This expanded into backend engineering via Firebase Authentication and Firestore, with server-side API routes handling plan generation, usage enforcement, and persistence.

The most meaningful growth came in applied data reasoning: learning to aggregate sparse, noisy health data; detect week-over-week trends; handle missing values gracefully; and design schemas that tolerate partial input without failure. Exposure to AI systems shaped the core architectural decision: rule-based logic as the safety foundation, AI as an optional enhancement layer — never a dependency.

4. System Architecture Overview

Sweeten is a modular Next.js 15 web application. The architecture separates presentation, data storage, and plan-generation layers for maintainability, auditability, and independent failure handling.

4.1 Frontend Layer

Users input daily vitals via a calendar-style interface. Blood glucose is required; insulin dosage, carbohydrate intake, weight, steps, mood, and notes are optional. The dashboard shows weekly summaries and the most recent generated plan. UX design prioritizes clarity, soft framing, and low cognitive load — treating daily logging as a moment of self-care rather than clinical obligation.

4.2 Data Model

Vitals are stored per-user, per-day using a deterministic Firestore document path (`users/{uid}/vitals/YYYY-MM-DD`). This schema intentionally supports partial and sparse data — a key aspect of the research problem — enabling the system to function with as little as one week of blood glucose readings.

4.3 Plan Generation Pipeline

The core research contribution is a two-tier plan-generation architecture:

- **AI-Assisted Generation:** Server-side API routes invoke Google Gemini (`gemini-2.5-flash`). Enforced constraints include one plan per user per week and regeneration only on significant vitals change. AI is never invoked client-side.
- **Deterministic Fallback Engine** (`planEngine.ts`): Automatically activates on AI failure. Aggregates vitals, computes trends, and generates structured weekly plans using rule-based logic — producing the same output schema as AI-generated plans.

4.4 Shared Analytics Layer

Both pipelines share a common analytics module (`planAnalytics.ts`) that computes average glucose, week-over-week trends, activity changes, and data completeness indicators. This shared foundation enables direct, controlled comparison between AI-generated and deterministic plans — a key evaluation mechanism.

5. Code-Level Explanation of the Plan Engine

Illustrative Example

Input: avg glucose = 165 mg/dL, avg steps = 3,800/day → Classification: elevated glucose + low activity → Focus: Activity Rebuild Week with Glucose Stability emphasis → Output: light post-meal aerobic activity, gradual step increase, carbohydrate awareness

5.1 Data Aggregation and Validation

The last 7 days of vitals are retrieved and normalized. Blood glucose is the primary signal; all other parameters act as modifiers. Missing values are explicitly handled with documented fallback behavior — the system degrades gracefully rather than erroring.

5.2 Threshold-Based Classification

Glucose readings are categorized as hypoglycemic, target-range, or hyperglycemic using ADA clinical thresholds. Comparative logic detects sustained trends across the week rather than reacting to isolated outliers — producing proportional, non-reactive recommendations.

5.3 Plan Synthesis

The deterministic engine assembles structured plans using conditional templates mapped to plan-focus types. The optional AI layer (Gemini) refines phrasing, sequencing, and tone while the core logic structure remains unchanged. This separation ensures the safety-critical decision layer is always auditable.

6. Research Basis: Guideline-to-Logic Traceability

Every threshold in Sweeten's plan engine maps to a published clinical guideline. This traceability — from ADA recommendation to conditional logic — is an explicit design goal, ensuring the system's decisions can be explained and audited.

ADA Guideline	Threshold / Trigger	Plan Focus
Postprandial glucose target	avg BG > 180 mg/dL	Glucose Stability Focus Week
Physical activity (≥150 min/wk)	< 5,000 steps/day	Activity Rebuild Week
Medication consistency	insulin consistency score	Adherence & Routine Week
Glycemic variability management	high SD in BG readings	Dietary Carb Awareness Week

Supporting Principles

- Trend sensitivity: sustained patterns weighted over isolated outliers
- Physical activity: moderate exercise shown to improve insulin sensitivity
- Carbohydrate awareness: manages glycemic variability without aggressive restriction
- Behavioral framing: non-punitive, achievable recommendations per BCW framework

7. Innovation and Contribution

Sweeten's primary innovation is its **failure-first, deterministic-core architecture**. Rather than building around AI and adding fallbacks as an afterthought, Sweeten treats the deterministic engine as the primary system and AI as an optional enhancement. This inversion has meaningful consequences for safety, explainability, and deployment in low-resource environments.

The shared analytics layer is a secondary contribution: because both pipelines consume identical computed metrics, Sweeten enables controlled, reproducible comparison of AI-generated and rule-based health plans — a capability most hybrid systems lack.

8. Methodology and Evaluation Strategy

Sweeten is evaluated as a software research system, not a medical treatment. This distinction shapes the evaluation methodology:

- **Robustness testing:** AI availability simulated as present, degraded, and absent — verifying fallback activation and output consistency.
- **Behavioral consistency:** AI-generated and deterministic plans compared on structure, focus selection, and recommendation categories using the shared analytics layer.
- **Failure simulation:** missing data, extreme glucose values, and zero-entry weeks tested for graceful degradation rather than silent errors.
- **Qualitative analysis:** plan completeness, tone appropriateness, and clinical alignment assessed against ADA guideline mappings.

This evaluation framework enables meaningful assessment without clinical trials, while surfacing the architectural questions the research is designed to answer.

9. Social Impact and Relevance

Pakistan has over 33 million adults living with diabetes, many in areas where specialist access is limited or nonexistent. Sweeten's hybrid architecture directly addresses this context: the system provides structured, personalized guidance without requiring constant internet connectivity, expensive devices, or clinical supervision.

The behavioral framing — gentle, non-punitive, habit-focused — is itself a design research decision, grounded in the Behaviour Change Wheel framework. Making daily health logging feel supportive rather than clinical increases long-term engagement, which is the prerequisite for meaningful data collection and plan personalization.

10. Limitations and Ethical Considerations

System Constraints

- Not a medical device; not for clinical use
- No diagnostic or medication dosing logic
- AI layer is optional and safety-constrained
- No real-time or emergency intervention logic

Privacy and Safety Controls

- Data privacy enforced via Firebase security rules
- Weekly plan limits prevent over-generation
- AI never invoked client-side; all calls server-gated
- No personal health data used for model training

11. Reflection: What I Learned

The most instructive finding was that deterministic and AI-generated plans agreed on plan focus selection in the majority of tested cases — but diverged meaningfully on tone, sequencing, and specificity of recommendations. This suggests the AI layer's primary value is not correctness but communication quality, which has implications for how hybrid health systems should be evaluated.

The hardest engineering problem was not the AI integration but the analytics layer: designing aggregation logic that handles partial weeks, missing modalities, and extreme outliers without producing misleading summaries. Real-world health data is messier than synthetic test cases reveal.

Future work would benefit from clinician review of plan outputs and localized adaptation of ADA thresholds for South Asian metabolic profiles, which differ from the Western populations on which most guidelines are based.

12. Conclusion and Future Scope

Sweeten is a technically grounded, research-oriented exploration of hybrid intelligence in digital health. It demonstrates that a health-planning system can remain safe, personalized, and interpretable without continuous AI availability — and provides a reference architecture for doing so responsibly.

Future Directions

- Clinician-validated evaluation of plan clinical alignment
- Localized guideline adaptation for South Asian metabolic profiles
- Integration with glucometers and consumer wearables
- Formal ablation study: AI vs. deterministic plan quality metrics
- Sustainable deployment via public health or NGO partnerships
- Extension of architecture to other chronic condition domains

13. Jury Summary

Sweeten is a solo-developed research system investigating how hybrid AI and deterministic architectures can deliver safe, reliable diabetes self-management support under real-world constraints. The system collects daily vitals and generates weekly personalized plans via a two-tier pipeline: a Gemini-powered AI layer for enhanced personalization, and a rule-based fallback engine for guaranteed reliability. A shared analytics layer enables direct comparison of both outputs. Key strengths: explicit ADA guideline-to-logic traceability, failure-first design philosophy, ethical AI constraints, and relevance to low-resource deployment contexts. Not a medical device; functions as a decision-support and habit-structuring tool.

14. Solo Participant Justification

This project was entirely designed, implemented, and evaluated by a single developer-researcher. The scope was chosen deliberately to demonstrate depth across frontend engineering, backend architecture, data modeling, clinical reasoning, and rule-based system design — disciplines that are rarely combined in a single student project. Solo development enabled cohesive architecture, rapid design iteration, and direct ownership of every research and technical decision.

15. References

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